6	**(a)**	**(i)**	$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$. Symbol, mass number and atomic number of alpha particle Symbol, mass number and atomic number of proton Correct representation of nuclear equation	
		(ii)	Sum of the masses of $^{14}_{7}N$ and an α-particle = 13.9993 u + 4.0015 u = 18.0008 u Sum of the masses of $^{17}_{8}O$ and proton = 16.9947 u + 1.0073 u = 18.0020 u increase in mass during reaction = 18.0020 u - 18.0008 u = 0.0012 u The minimum energy is to be supplied to the reactants in the form of kinetic energy of the α-particle The kinetic energy of the alpha particle = mass increase x c^2 = 0.0012 u x c^2 = $0.0012 \times 1.66 \times 10^{-27} \times (3 \times 10^{8})^2$ = 1.79×10^{-13} J	
	(b)	A radioactive isotope of thallium $^{207}_{81}Tl$ emits a *β*-particle and is thought to emit a gamma photon. The half-life of $^{207}_{81}Tl$ is 135 days.		
	(b)	**(i)**	Circular paths of small radii show that beta particles are present and that beta particles are charged. Circular paths of different radii show that the beta particles have different speed and hence different kinetic energy. The faint straight line path shows the presence of gamma ray photons which have no charge and not deflected by the magnetic field. The faint path shows that the gamma ray photon has much less ionization ability as compared to the beta particles. Max 3 marks	
		(ii)	Energy available from 1 g of thallium-207 = number of thallium-207 atoms x 2.4×10^{-13} = $(1/207) \times 6.02 \times 10^{23} \times 2.4 \times 10^{-13}$ = 6.98×10^{8} J	
		1.	Initial rate of β-particle emission, A_o = (initial) activity of thallium = decay constant x number of thallium nuclei = (ln 2 / half-life) x number of thallium nuclei = $(\ln 2 / 135) \times (1/207) \times 6.02 \times 10^{23}$ = 1.493×10^{19} = 1.5×10^{19} day^{-1}	
		2.	Initial power = initial rate x energy of a β-particle = $[1.493 \times 10^{19} / (24 \times 60 \times 60)] \times 2.4 \times 10^{-13}$ = 41.5 W	

Section B

7	(a)	(i)	The rod is in vertical equilibrium only if Tension is acting vertically upwards, since	